OBD2 sniffer: revision A

A read-only CAN and OBD2 scanner. Powered from the laptop's USB port only.
Design notes and requirements: docs/design/circuit-board.md, docs/design/requirements.md.
Parts, datasheets and the pre-order checklist: hardware/rev-a/README.md.

Sheets

USB and power



File: usb-power.kicad_sch

USB-C socket, fuse, ESD protection,
3.3 V regulator, PWR light.

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File: obd2-sniffer-rev-a.kicad_sch

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USB and power

The board is powered from the laptop's USB port only (PWR-5).
The car's 12 V supply has no connection to the board (PWR-8).

USB-C socket

5.1 k on each CC pin: the host switches its 5 V on.

Fuse

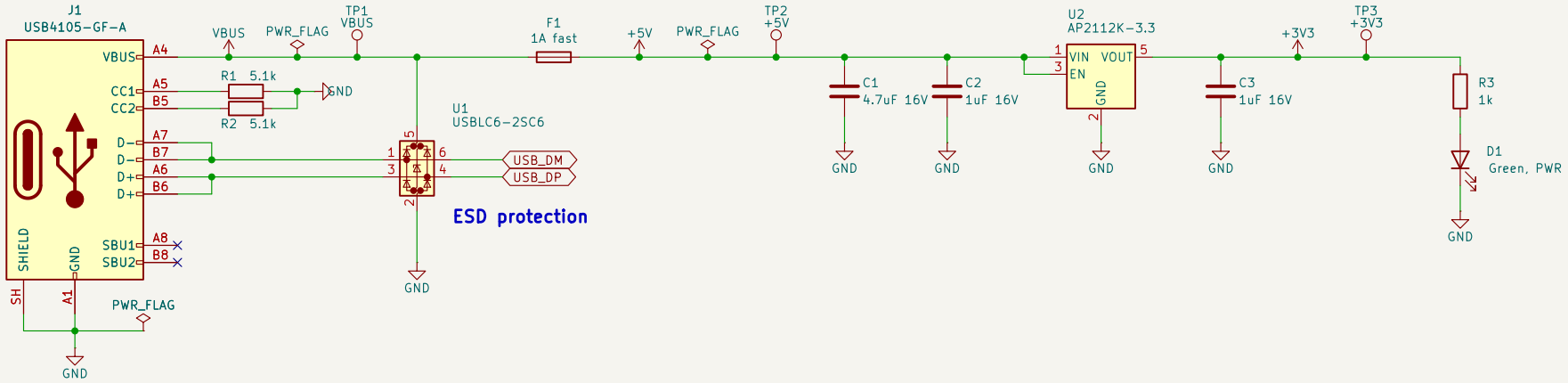
1 A fast, 0.13 ohm: a short on the board can't draw a dangerous current.

3.3 V regulator

1 uF ceramic in and out, as the datasheet asks. Enable is tied to the input: always on.

PWR light

On whenever 3.3 V is present, with or without firmware (USE-3).



The socket's shield is tied to ground. SBU pins are not used.

Total capacitance on the USB 5 V stays under 10 uF, the limit a USB device may present when plugged in.
The transceivers take the 5 V directly and need at least 4.5 V: the fuse drops about 35 mV at 250 mA.

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